

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- ANALYTICAL PROBLEM SOLVING

Course Code:	CSA1205	Course Title:	Computer Architecture
CO Assessed:	CO2	Assessment:	ASSESSMENT TOOL1- ANALYTICAL PROBLEM SOLVING
Weightage:	45%	Total Marks:	25
CO2 Statement:	Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems. (BL3)		
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ANNEXURE A

1. A processor performs arithmetic operations on signed 8-bit integers using 2's complement representation. Consider two numbers: +45 and -23. Analyze in detail how the system performs both addition and subtraction, including binary conversion, alignment of sign bits, and intermediate steps. Determine whether overflow occurs, and explain how the processor detects overflow and interprets the final result.
2. A high-performance computing system replaces a ripple carry adder with a 4-bit carry look ahead adder (CLA) to reduce computation delay. Given two binary inputs, analyze how generate and propagate signals are formed and how carries are computed in advance. Compare the time delay of CLA with ripple carry addition and justify, with reasoning, why CLA improves speed in modern processors.
3. In a signed multiplication operation, a processor uses Booth's algorithm to multiply two numbers, -6 and +5. Analyze the algorithm step-by-step, including the role of the accumulator, multiplier, and Booth bit, along with arithmetic shifts. Explain how the algorithm efficiently handles signed numbers and reduces the number of addition/subtraction operations compared to traditional multiplication.

4. A processor is required to perform division of two binary numbers (e.g., $13 \div 3$) using both restoring and non-restoring division algorithms. Analyze both methods step-by-step, showing intermediate remainders and quotient formation. Compare the two approaches in terms of number of steps, complexity, and efficiency, and explain why non-restoring division is often preferred in modern systems.

5. A scientific application requires precise handling of real numbers using the IEEE 754 standard for floating-point representation. Given a decimal number (e.g., -13.25), analyze how it is represented in both single precision and double precision formats, including sign, exponent, and mantissa fields. Further, explain how floating-point addition is performed between two such numbers, highlighting issues like normalization, rounding, and precision loss.

Code of Conduct:

*I **DEEPA PREYA H/192421047** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

MARKS ALLOCATION

	Problem Understanding (20%)	Method Selection (20%)	Solution Process (25%)	Accuracy of Calculation (20%)	Interpretation of Results (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with	Partial understanding; misses key	Misinterprets or unable to

		requirements accurately	minor gaps	elements	understand problem
Method Selection	20%	Selects most appropriate method/ formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q1. Signed 8-bit 2's Complement Addition and Subtraction (+45 and -23)

Step 1: Convert Decimal to 8-bit Binary

+45

- Binary = 00101101

-23

- +23 = 00010111
- 1's complement = 11101000
- 2's complement = 11101001

A) Addition: (+45) + (-23)

00101101 (+45)

+ 11101001 (-23)

00010110

Result: 00010110 = +22

Final Answer: +22

B) Subtraction: (+45) - (-23)

Subtraction is performed as:

(+45) + (+23)

00101101 (+45)

+ 00010111 (+23)

01000100

Result: 01000100 = +68

Q2. 4-bit Carry Look Ahead Adder (CLA)

Given Inputs

- A = 1011
- B = 0110

Step 1: Find Generate (G) and Propagate (P)

Formula:

Bit	A	B	$G = A \cdot B$	$P = A \oplus B$
0	1	0	0	1
1	1	1	1	0
2	0	1	0	1
3	1	0	0	1

Step 2: Compute Carry Signals

Assume initial carry $C_0 = 0$

- $C_1 = G_0 + P_0C_0 = 0$
- $C_2 = G_1 + P_1C_1 = 1$
- $C_3 = G_2 + P_2C_2 = 1$
- $C_4 = G_3 + P_3C_3 = 1$

Step 3: Find Sum Bits

Formula:

$$S_i = P_i \oplus C_i$$

- $S_0 = 1 \oplus 0 = 1$
- $S_1 = 0 \oplus 0 = 0$
- $S_2 = 1 \oplus 1 = 0$
- $S_3 = 1 \oplus 1 = 0$

Final Sum = $C_4S_3S_2S_1S_0 = 10001_2$

Q3. Booth's Algorithm: Multiply $-6 \times +5$

Step 1: Convert Numbers to 4-bit 2's Complement

- Multiplicand (M) = $-6 = 1010$
- $-M = +6 = 0110$
- Multiplier (Q) = $+5 = 0101$

Initial values:

- A = 0000 (Accumulator)
- Q = 0101 (Multiplier)
- $Q_{-1} = 0$
- Count = 4

Step 2: Booth's Algorithm Table

Step	$Q_0 Q_{-1}$	Operation	A	Q	Q_{-1}
Initial	—	Initial	0000	0101	0

1	10	$A = A - M$	0110	0101	0
		Arithmetic Right Shift	0011	0010	1
2	01	$A = A + M$	1101	0010	1
		Arithmetic Right Shift	1110	1001	0
3	10	$A = A - M$	0100	1001	0
		Arithmetic Right Shift	0010	0100	1
4	01	$A = A + M$	1100	0100	1
		Arithmetic Right Shift	1110	0010	0

Step 3: Final Result

Final AQ = 11100010

This is an 8-bit 2's complement number.

Convert to decimal:

- $11100010_2 = -30$

Final Answer:

$$-6 \times +5 = -30$$

Q4. Restoring and Non-Restoring Division ($13 \div 3$)

Given

- Dividend = $13 = 1101_2$
- Divisor = $3 = 0011_2$

A) Restoring Division

Initial Values

- $A = 0000$
- $Q = 1101$
- $M = 0011$
- $\text{Count} = 4$

Step	Operation	A	Q
Initial	Initial	0000	1101
1	Shift left, $A = A - M$ (negative, restore)	0000	1010
2	Shift left, $A = A - M$ (negative, restore)	0001	0100
3	Shift left, $A = A - M$ (positive, $Q_0 = 1$)	0000	1001
4	Shift left, $A = A - M$ (positive, $Q_0 = 0$)	0001	0100

Final Result

- Quotient (Q) = $0100 = 4$
- Remainder (A) = $0001 = 1$

B) Non-Restoring Division

Initial Values

- $A = 0000$
- $Q = 1101$
- $M = 0011$

Step	Operation	A	Q
Initial	Initial	0000	1101
1	Shift, subtract M	1101	1010
2	Shift, add M	0000	0100
3	Shift, subtract M	1101	1001
4	Shift, add M	0001	0100

Final Result

- Quotient (Q) = $0100 = 4$
- Remainder (A) = $0001 = 1$

Q5. IEEE 754 Floating-Point Representation of -13.25

Step 1: Convert Decimal to Binary

- $13 = 1101_2$
- $0.25 = 0.01_2$

Therefore,

$$13.25 = 1101.01_2$$

Normalize:

$$1101.01_2 = 1.10101 \times 2^3$$

A) IEEE 754 Single Precision (32-bit)

Fields

- Sign bit (S) = 1 (negative number)
- Exponent (E) = $3 + 127 = 130 = 10000010_2$
- Mantissa (M) = 101010000000000000000000

Sign	Exponent	Mantissa
1	10000010	101010000000000000000000

32-bit Representation:

1 10000010 101010000000000000000000

B) IEEE 754 Double Precision (64-bit)

Fields

- Sign bit (S) = 1
- Exponent (E) = $3 + 1023 = 1026 = 10000000010_2$
- Mantissa (M) = 10101000000000000000000000000000000000000000

Sign	Exponent	Mantissa
1	10000000010	10101000000000000000000000000000000000000000

64-bit Representation:

1 10000000010 10101000000000000000000000000000000000000000